Assignment 3

Vibration of Structures (ME60428)
Spring 2025-26
40 marks

1. A sliding–fixed steel string of length l and having a circular cross-section of 5 mm diameter, is excited by a uniform harmonic force $q(x,t) = Q_0 \cos \Omega t$, as shown in Figure 1. The k-th eigenvalue and eigenfunction of this system is given by $\omega_k = (2k-1)\frac{\pi c}{2l}$ and $W_k(x) = \cos{(2k-1)}\frac{\pi x}{2l}$, respectively. Answer the following questions:

   (a) Assuming the steady-state response as $w_p(x,t) = X(x)\cos\Omega t$, obtain the boundary value problem (BVP) of the system, including the boundary terms.

   (b) Analytically determine the steady-state response $w_p^A(x,t)$ of the system

   (c) Using the eigenfunction expansion method $X(x) = \sum\limits_{k=1}^{\infty} \alpha_k W_k(x)$, determine the steady-state response $w_p^E(x,t)$ of the system.

   (d) Derive the Green's function $G(x, \bar{x}, \Omega)$ for the system, and use it to compute the steady-state response $w_p^G(x,t)$ of the system using the Green's function method.

   (e) Compare the shape of the string obtained using the three methods [$w_p^A(x,t)$, $w_p^E(x,t)$ & $w_p^G(x,t)$ vs x] for any three different time instants of your choice and show the results in three separate graphs [use 10-term expansion when plotting $w_p^E(x,t)$]

Take $l = 2$ m, $\rho = 7850$ kg/m^3, $T = 1000$ N, $Q_0 = 60$ N and $\Omega = 220$ rad/s

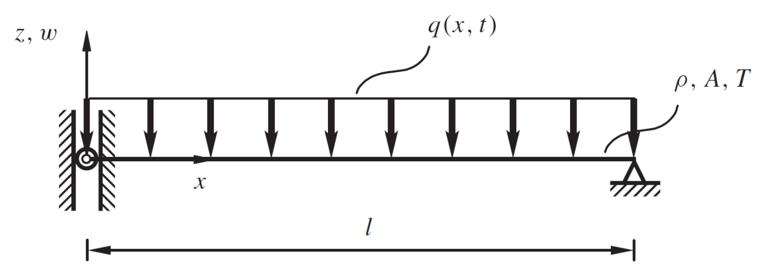

Figure 1: Sliding-fixed string with uniform harmonic load

Instructions

- All derivations and necessary steps should be hand-written. Any assisting codes and/or plots must be attached with the written assignment
- Scan and upload the complete assignment as a **single PDF file**. All handwritten pages must clearly display your name, roll number, and physical signature at the top-right corner of each page.
- The assignment should be presented in a clear, concise, and well-organized manner, with legible handwriting and appropriate cleanliness and neatness throughout.
- You may discuss the assignment with your peers to clarify concepts; however, **plagiarism in any form is strictly prohibited**, and the submitted work must be entirely your own.